BHUVANA NAGARAJ

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Professional Summary

AI Engineer with **6 years** of production experience deploying **ML models and inference pipelines** at global scale across **12M+ devices**. Proven expertise in **computer vision, LLMs, on-device AI**, and **explainability (XAI)** using TensorFlow, PyTorch, and OpenCV. Built and shipped AI systems across Samsung's flagship device ecosystem spanning **camera, ads, and multimedia** domains. Graduated in **M.S. in Computer Science** (AI/ML focus, GPA 3.70) from Northern Arizona University.

Technical Skills

- **AI/ML :** TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, CNNs, FNNs, RNNs, LSTMs, GANs, Diffusion Models, XAI (SHAP, LIME)
- **LLMs & GenAI :** GPT-4, Claude, LLaMA, Prompt Engineering, RAG, LangChain, LlamaIndex, Fine-tuning, RLHF, Oracle GenAI (certified)
- **Computer Vision :** OpenCV, PIL, YOLO, Image Segmentation, Object Detection.
- **NLP :** Transformers, BERT, T5, GPT, Sentence Embeddings, Named Entity Recognition, Sentiment Analysis, Text Classification
- **Languages :** Python, Java, C, C++, SQL, JavaScript, Shell/Bash
- **Data & Cloud :** AWS (EC2, S3, Lambda, SageMaker), GCP (BigQuery, Pub/Sub), Apache Kafka, Spark, ETL pipelines
- **Databases :** SQL, NoSQL, Cassandra, Elasticsearch, Redis, Logstash, Kibana
- **Dev & MLOps :** Docker, Kubernetes, CI/CD, Git, Perforce, MLflow, Weights & Biases, Flask, FastAPI, Protocol Buffers, Tableau

Certifications: OCI Associate Architect, Oracle GenAI Professional, Oracle Cloud Professional

Experience

Easley-Dunn Productions, Inc. — **Mar 2025 – Present**
AI Engineer — *San Francisco, CA*

- Architected **LLM-powered NPC dialogue systems** using **RAG** enabling context-aware interactions across global player base.
- Built **reinforcement learning agents** for adaptive gameplay deployed across **multi-region production environments**.
- Optimized **GenAI inference pipelines** for procedural content generation reducing asset creation time **40% worldwide**.
- Designed **real-time ML behavior trees** enabling intelligent NPC decision-making at scale across diverse game worlds.
- Integrated **multimodal AI models** for dynamic audio-visual game content personalized across international player segments.
- Deployed **on-device ML inference** optimizing game AI performance across globally distributed hardware platforms.

Samsung Research — **Jul 2018 – Jan 2024**
Senior Associate Engineer — *Bangalore, India*

Camera Systems
- Deployed **TensorFlow/OpenCV CNN models** for real-time defect detection boosting accuracy **30% across 195+ countries**.
- Architected **on-device ML inference framework** achieving **34% faster capture latency** across **12M+ devices worldwide**.
- Shipped **ML-driven image quality pipelines** across Galaxy S/Note/Tab/Foldables ensuring consistent AI performance globally.
- Trained and optimized **CNN architectures** for on-device visual analysis deployed across **international flagship lineups**.
- Designed **ML-based sensor calibration pipelines** ensuring imaging consistency across **globally diverse SoC variants**.
- Delivered **camera AI features** for Galaxy S21 milestone ahead of schedule impacting **millions of users worldwide**.

Samsung Digital Ads
- Engineered **ACR-based ML ad-validation pipeline** increasing detection precision **30% across global ad inventory**.
- Integrated **ML inference** within **Jenkins CI/CD** doubling release velocity across **internationally distributed teams**.
- Implemented **Kafka real-time ML fraud detection** processing **billions of ad events** under **sub-50ms** latency globally.
- Deployed **Python/Java microservices** automating ML ad-verification reducing operational costs **20% companywide**.
- Delivered end-to-end **AI-driven ads tracking** pipeline for **U.S. market** earning Samsung Spot Award recognition.
- Scaled **ML-powered ad attribution** systems supporting **multi-region campaigns** across Samsung's global device ecosystem.

Multimedia & On-Device AI
- Shipped **TensorFlow Lite** modules achieving **2x faster ML inference** across **globally distributed device ecosystems**.
- Benchmarked **ML model throughput** on **Snapdragon SoCs** validating performance across **international hardware variants**.
- Delivered **ML pipeline validation suite** reducing deployment crash rates **40%** across **global release pipelines**.
- Optimized **encoder-decoder ML pipelines** improving frame latency **35%** across cross-functional international teams.
- Contributed **multimedia SDKs** integrating third-party ML plug-ins expanding AI capabilities across global device ecosystem.
- Co-developed **cross-platform AI multimedia frameworks** enabling consistent ML performance across **diverse global markets**.

Projects

Explainable AI for Autonomous Cybersecurity Threat Detection | *TensorFlow, SMOTE, SHAP* **2025**

- Designed **FNN-based IoT threat pipeline** classifying **28 cyberattack types** under severe class imbalance.
- Applied **SMOTE and focal loss** on **2.5M+ IoT-23 samples** achieving **76% detection accuracy** in TensorFlow.
- Integrated **SHAP explainability** generating per-prediction attributions for transparent, audit-ready AI decisions.

AutoBugCam – Real-Time Camera Artifact Detector | *PyTorch, UNet, ONNX, OpenCV* **2024**

- Built **dual-head segmentation model** detecting HDR halos, moire, flare, and **8 camera artifacts** per-pixel.
- Designed **synthetic artifact generators** producing supervised masks for controlled **ML training pipelines**.
- Exported trained **PyTorch model to ONNX** enabling cross-platform deployment and **mobile inference experiments**.

Unsupervised IoT Anomaly Detection | *PyTorch, Autoencoder, SHAP, Scikit-learn* **2024**

- Built **deep autoencoder** for unsupervised IoT anomaly detection using **reconstruction error thresholding**.
- Achieved high **ROC-AUC and PR-AUC** benchmarked against Isolation Forest and **One-Class SVM baselines**.
- Designed **config-driven reproducible ML pipeline** with robust preprocessing, PCA, and **SHAP attribution**.

DuoCoder-NLP – Two-Agent MCP Code Generation System | *Python, NLP, MCP, Transformers* **2025**

- Architected **dual-agent MCP system** with Spec-Planner and Builder-Verifier agents for **autonomous code generation**.
- Implemented **NLP-driven ticket parsing** generating acceptance criteria, pytest stubs, and **implementation plans**.
- Integrated **MCP tool orchestration** (fs, git, process, http) enabling **end-to-end automated ML experimentation**.

Education

Northern Arizona University **Dec 2025**
Master of Science (MS) in Computer Science

- **Coursework:** Data Structures, Algorithms, High Performance Computing, Systems Software, Databases, Machine Learning, Deep Learning, Artificial Intelligence, Large Language Models

Recognitions

- **Samsung Spot Award (2023):** End-to-end AI-driven ads tracking and ML validation pipeline for U.S. region.
- **Samsung Spot Award (2021):** Early delivery of Galaxy S21 AI camera milestones boosting imaging throughput 34%.
- **Anthropic AI Hackathon Finalist (SF):** Built scalable explainable ML system using MCP architecture.
- **MCP Hackathon Participant (SF):** Developed autonomous multi-agent AI pipeline for real-time ad validation.
- **Oracle Certified:** OCI Associate Architect, Oracle GenAI Professional, Oracle Cloud Professional.
- **Academic Excellence:** GPA 3.70/4.00 – M.S. Computer Science (AI/ML), Northern Arizona University.